FRAGEN MASTERPRÜFUNG

Folgende Liste ist eine Sammlung von Fragen für die Masterprüfung für die Mastervorlesungen "Stochastic Analysis" und "Selected Topics in Probability Theory". Sie ist keinesfalls "vollständig" und darf gerne weiter ergänzt werden. Die Vorlesung basiert auf [1]. Zu den Übungsaufgaben im Buch von Le Gall gibt es auch Lösungen, siehe [2].

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Bemerkung: Diese Fragensliste ist unvollständig und befindet sich in Arbeit.

Part 1. Stochastic Analysis

1. Gaussian spaces and Gaussian processes

Question 1. Suppose that $X_n \sim \mathcal{N}(m_n, \sigma_n^2)$ and $X_n \xrightarrow[n\to+\infty]{d} X$. What can you say about X? Are there other modes of convergence that hold in this case?

Question 2. What is a Gaussian space? What is a Gaussian process?

Question 3. Let S be a d-dimensional Euclidean space and let $\gamma : S \to S$ be a linear map which is positive semi-definite and symmetric, that is,

$$(u, \gamma u)_S \geq 0 \quad \text{and} \quad (u, \gamma v)_S = (\gamma u, v)_S \quad \text{for all } u, v \in S.$$

Prove that for any such γ there exists a Gaussian vector $\boldsymbol{X}$ on S such that

$$\mathbb{V}\mathrm{ar}\big((u, \boldsymbol{X})_S\big) = (u, \gamma u)_S. \tag{1}$$

Then prove the opposite (and much easier) direction: Given an S-valued Gaussian vector $\boldsymbol{X}$, prove that there exists some $\gamma_{\boldsymbol{X}} = \gamma : S \to S$ linear such that (1) holds.

Question 4. Consider the setting from Question 3. Suppose that $\gamma = \gamma_{\boldsymbol{X}}$ is diagonal in some orthonormal basis $e_1, \dots e_d$ with $\gamma(e_i) = \lambda_i e_i$ with $\lambda_1 \geq \dots \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_d$. Show that if we define Y_i, $i = 1, \dots, d$ by

$$\boldsymbol{X} = \sum_{i=1}^{r} e_i Y_i, \quad \text{that is} \quad Y_i = (\boldsymbol{X}, e_i)_S$$

then Y_i are independent with $Y_i \sim \mathcal{N}(0, \lambda_i)$. Show that $d = r$ if and only if $\boldsymbol{X}$ has density

$$p_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d \cdot \det \gamma_{\boldsymbol{X}}}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}, \gamma_{\boldsymbol{X}}^{-1}\boldsymbol{x})_S\right\}, \quad \boldsymbol{x} \in S.$$

(You also have to show that in this case, γ is invertible).
Hint: In order to show the independence of Y_i prove and use the fact that the components of a Gaussian vector $\boldsymbol{X} = [X_i]_{i=1}^d$ (in $\mathbb{R}^d$) are independent if and only if the covariance matrix $\Sigma = [\mathrm{Cov}(X_i, X_j)]_{i,j=1}^d \in \mathbb{R}^{d\times d}$ is diagonal. You may use the characteristic function for this.

Question 5. Explain and prove the connection between subspaces of a Gaussian space and the σ-algebras generated by them (orthogonality).

Question 6. Show Corollary 3.5 in the notes.

Question 7. What is Gaussian white noise? Give the definition of it and prove that such objects exist. What can you say about its "quadratic variation" and how is it connected to the variation of Brownian motion?

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2. Brownian Motion

Question 8. What is (Pre)-Brownian Motion. What are equivalent definitions of it? What is the finite-dimensional distribution of Brownian motion?

Question 9. What do we need to go from Pre-Brownian motion to Brownian motion? Give the proof of Kolmogorov's Continuity theorem. Use it to deduce the existence of Brownian motion.

Question 10. What is the *Wiener measure*? On which space is it defined. What does Brownian motion look like on this space?
Show that the Wiener measure is uniquely defined on said space.

Question 11. What is the quadratic variation of a function? Give and prove a formula for the quadratic variation of a Brownian motion B. Prove that the (non-quadratic) variation of Brownian motion is a.s. infinite.

Question 12. What is a filtration? When is it right-continuous? How can you make it right-continuous? What is the natural filtration of Brownian Motion?

Question 13. State and prove the simple Markov-property of Brownian Motion. What are its corollaries?

Question 14. What is a stopping time? How do stopping times in discrete and continuous time differ? Give an example of a stopping time. Explain why the entrance of an open set is not a stopping time. Define the σ-algebra $\mathcal{A}_T$ for a stopping time T and explain its meaning and show that T is $\mathcal{A}_T$ measurable. Prove that an $\{\mathcal{A}_t\}_t$-stopping time is also an $\{\mathcal{A}_t^+\}_t$-stopping time.

Question 15. Let $\{\mathcal{F}_t\}_{t\geq 0}$ be the natural filtration of Brownian Motion, i.e., $\mathcal{F}_t = \sigma(B_s; s \leq t)$ and let T be a stopping time. Show that B_T is a measurable map.

Question 16. State and show the strong Markov-property of Brownian motion.

Question 17. State, prove and explain the reflection principle of Brownian motion. Also explain the connection between $\sup_{s\leq t} B_s$ and $|B_t|$.

Question 18. What are the *usual conditions*? Why are they important? Show that the completion of the measure space with respect to the Wiener measure and the completion of the natural filtration of Brownian motion satisfy the usual conditions.

Question 19. In which ways can stochastic processes be measurable? Why is this relevant. Do those definitions imply each other? What is the progressive σ-algebra $\mathcal{A}_{\text{prog}}$?

Question 20. Let $\{X_t\}_{t\geq 0}$ be a progressively measurable process taking values in $(S, \mathcal{S})$ and let T be a stopping time. Then the function

$$X_T(\omega) = \begin{cases} X_{T(\omega)}(\omega), & \text{if } T(\omega) < +\infty \\ x_0 \in S, & \text{if } T(\omega) = +\infty \end{cases}.$$

Then show that if $A \subseteq S$ is closed,

$$H_A := \inf \{t \geq 0 \mid X_t \in A\}$$

is a $\{\mathcal{A}_t\}_t$-stopping time and if $A \subseteq S$ is open, then H_A is a $\{\mathcal{A}_t^+\}_t$-stopping time. (Note this last part was not done in the lecture).

3. Martingales

Question 21. Give the definition of a martingale (time continuous). Why are martingales important? What are sub-martingales and super-martingales?
Give an example of a martingale. What martingales can you "build" using Brownian motion.

Question 22. What is the Wiener-integral? What martingales can you "build" using the Wiener integral?

Question 23. State and prove Doob's inequality. For the proof you may use Doob's inequality for discrete time martingales?

Question 24. What can you say about the convergence of (sub/super-) martingales? What are Upcrossings? What is the Upcrossing inequality? What does it have to do with the convergence of martingales?
Hint: To answer this question you need that for a sub/super-martingale $\{X_t\}_{t\geq 0}$ that

$$\sup_{s\in[0,t]} \mathbb{E}[|X_s|], \quad \text{for all } t \geq 0,$$

and the so-called Backward sub-martingale convergence theorem.

Question 25. Using the results of Question 24 show that the following theorem holds:

Theorem. *Suppose that the filtration $\{\mathcal{A}_t\}_{t\geq 0}$ satisfies the usual conditions, $\{X_t\}_{t\geq 0}$ is a sub-martingale and that the map $0 \leq t \mapsto \mathbb{E}[X_t]$ is right continuous. Then X has a modification Y that has càdlàg trajectories and is also an $\{\mathcal{A}_t\}_{t\geq 0}$ sub-martingale.*

Question 26. Which martingales do have a limit for $t \to +\infty$? In which sense does the martingale converge to that limit?

Question 27. What are closed martingales? How do they relate to the concept of uniform integrability and convergence as $t \to +\infty$. What can you say about the limit X_∞?

Question 28. What does the optimal stopping theorem say? How do you prove it? Why is it important that one approximates the stopping time T with stopping times T_n that increase to T?

Question 29. State and prove some applications of the optimal stopping theorem.

Question 30. What is a local martingale? What are some basic properties of local martingales? Give an example of a local martingale that is not a martingale.

Question 31. Let $M = \{M_t\}_{t\geq 0}$ be a local martingale as well as a process of finite variation with $M_0 = 0$. Show that M is indistinguishable from 0.

Question 32. Let M be a local martingale. What is the Bracket process $\langle M, M\rangle$ of a semi-martingale? Give a sketch of the proof of the existence of the bracket process.

Question 33. Let M be a local martingale and T a stopping time. Show that

$$\langle M^T, M^T\rangle_t = \langle M\rangle_t^T := \langle M\rangle_{t\wedge T}.$$

Also show that

$$\langle M, M\rangle_t \equiv 0 \implies M_t \text{ is indistinguishable from } 0.$$

Question 34. Let M_0 be a local martingale with $M_0 \in L^2(\mathbb{P})$. Show that the following are equivalent:

- M is a true martingale bounded in $L^2$and
- $\mathbb{E}[\langle M\rangle_\infty] < +\infty$

and show that if these properties are satisfied, $M_t^2 - \langle M, M\rangle_t$ is a UI martingale.
Note that there are local martingales that are bounded in L^2 that are not true martingales. It is not enough to show boundedness in L^2.

4. Construction of the stochastic Integral

To aid with the memorization of the construction of the stochastic integral we lead through it with some question and give some information between those.

Question 35. Given a process $A = \{A_t\}_{t\geq 0}$ of finite variation with $A_0 = 0$ give the definition of

$$\int_{s=0}^{t} X_s \, \mathrm{d}A_s.$$

What properties does the process X have to satisfy that the integral is well defined?

Question 36. State and show the inequality of Kunita-Watanabe.
Hint: Note its similarity to Cauchy Schwartz.

4.1. Stochastic Integral with respect to martingales.

Define $\mathbb{H}^2$ as the space of all *continuous* martingales M with

$$\sup_{t\geq 0} \mathbb{E}[M_t^2] < +\infty, \quad \text{and} \quad M_0 \overset{\text{a.s.}}{=} 0.$$

Recall that this implies $\mathbb{E}[\langle M\rangle_\infty] < +\infty$ c.f. Question 34. Define

$$(M, N)_{\mathbb{H}^2} = \mathbb{E}[\langle M, N\rangle_\infty] = \mathbb{E}[M_\infty N_\infty].$$

Question 37. Show that $(\cdot,\cdot)_{\mathbb{H}^2}$ is a scalar product on $\mathbb{H}^2$. Show that $\mathbb{H}^2$ equipped with this scalar product is a Hilbert space.
Hint: Use Doob's inequality to show that

$$\mathbb{E}\left[\sup_{t\in[0,+\infty]} |M_t^n - M_t^m|^2\right] = 0$$

for any Cauchy-sequence $\{M^n\}_{n\in\mathbb{N}} \subseteq \mathbb{H}^2$.

We now need suitable integrands to integrate against. To this end we introduce the space $L^2(M)$ for $M \in \mathbb{H}^2$ by

$$L^2(M) := L^2(\mathbb{R}_+ \times \Omega, \mathcal{A}_{\text{prog}}, \mathrm{d}\mathbb{P}\, \mathrm{d}\langle M\rangle)$$

making $L^2(M)$ the space of all progressive processes H with

$$\mathbb{E}\left[\int_{\mathbb{R}_+} H^2 \, \mathrm{d}\langle M\rangle\right] < +\infty.$$

Like the Lebesgue integral we define the stochastic Integral on elementary/simple functions first and then extend to general functions. An *elementary process* is a process of the form

$$H_s(\omega) = \sum_{i=0}^{p-1} H^{(i)}(\omega)\mathbf{1}_{(t_i,t_{i+1}]}(s),$$

where $p \in \mathbb{N}$, $0 = t_0 < \cdots < t_p$ and $H^{(i)}$ is a bounded $\mathcal{A}_{t_i}$ measurable random variable. We denote the space of all such processes by $\mathcal{E}$.

Question 38. Given $M \in \mathbb{H}^2$ prove that $\mathcal{E}$ is a dense subspace of $L^2(M)$.
Hint: Use the following fact from functional analysis: Let H be a Hilbert space. A linear subspace $V \subseteq H$ is dense if and only if $h \perp V \implies h = 0$ for any $h \in H$.

References

[1] Jean-François Le Gall. *Brownian motion, martingales, and stochastic calculus* Graduate Texts in Mathematics, vol. 274, Springer [Cham], 2016. MR3497465

[2] Te-Chun Wang. *Solutions to Exercises on Le Galls Book: Brownian Motion, Martingales, and Stochastic Calculus.* Available at: `https://jupiter.math.nycu.edu.tw/~sheu/Solution.pdf`.

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